```python
import math

def calculate_azithromycin(total_dose, total_day, times_per_day, double_first_day=False):
    """
    Calculate azithromycin dosage for different formulations
    """
    print("\n=== Azithromycin Dosage Calculator ===")

    if double_first_day:
        print("NOTE: Patient needs double dose on first day")
        adjusted_total = total_dose * total_day + total_dice
        print(f"Total adjusted dose (including double first day): {adjusted_total}mg")
    else:
        adjusted_total = total_dose * total_day

    # Calculate for 100mg/5ml suspension
    n1_susp100 = math.ceil((total_dose * 5 / 100) * total_day / 15)
    n2_susp100 = math.ceil((total_dose * 5 / 100) * total_day / 30)
    q_susp100 = math.floor((total_dose * 5 / 100) / times_per_day)
    print("\nAzithromycin 100mg/5ml Suspension:")
    print(f"- Dose: {q_susp100}ml every {24/times_perday} hours")
    print(f"- Duration: {total_day} days")
    print(f"- Bottles needed: {n1_susp100} (15ml) OR {n2_susp100} (30ml)")

    # Calculate for 200mg/5ml suspension
    n1_susp200 = math.ceil((total_dose * 5 / 200) * total_day / 15)
    n2_susp200 = math.ceil((total_dose * 5 / 200) * total_day / 30)
    q_susp200 = math.floor((total_dose * 5 / 200) / times_per_day)
    print("\nAzithromycin 200mg/5ml Suspension:")
    print(f"- Dose: {q_susp200}ml every {24/times_perday} hours")
    print(f"- Duration: {total_day} days")
    print(f"- Bottles needed: {n1_susp200} (15ml) OR {n2_susp200} (30ml)")

    # Calculate for 250mg tablets
    n_tab250 = math.ceil((adjusted_total) / 250)
    q_tab250 = math.floor((total_dose / 250) / times_per_day)
    print("\nAzithromycin 250mg Tablets:")
    print(f"- Dose: {q_tab250} tablet(s) every {24/times_perday} hours")
    print(f"- Duration: {total_day} days")
    print(f"- Tablets needed: {n_tab250}")

    # Calculate for 500mg tablets
    n_tab500 = math.ceil((adjusted_total) / 500)
    q_tab500 = math.floor((total_dose / 500) / times_per_day)
    print("\nAzithromycin 500mg Tablets:")
```

```
    print(f"- Dose: {q_tab500} tablet(s) every {24/times_perday} hours")
    print(f"- Duration: {total_day} days")
    print(f"- Tablets needed: {n_tab500}")

def pertussis_treatment():
    """
    Main function to determine pertussis treatment
    """
    print("\n=== Pertussis Treatment Calculator ===")
    print("Please enter patient details:\n")

    try:
        # Get patient information
        weight = float(input("Patient weight (kg): "))

        # Handle age input with months/year conversion
        age_input = input("Patient age (months/years - e.g. '6 months' or '2 years'): ")
        if 'month' in age_input.lower():
            age_months = float(age_input.split()[0])
        else:
            age_months = float(age_input.split()[0]) * 12

        # Get treatment frequency
        times_per_day_input = input("\nEnter frequency (1-3), or 'default' for once daily:
").lower()
        times_per_day = 1 if times_per_day_input == 'default' else int(times_per_day_input)

        # Calculate total daily dose
        if age_months > 6:
            # For patients >6 months
            total_dose = 5 * weight
            if total_dose > 250:
                total_dose = 250
            double_first_day = True
        else:
            # For infants ≤6 months
            total_dose = 10 * weight
            if total_dose > 500:
                total_dose = 500
            double_first_day = False

        # Get treatment duration
        total_day_input = input("\nEnter treatment duration (3-7 days), or 'default' for 5 days:
").lower()
        total_day = 5 if total_day_input == 'default' else int(total_day_input)
```

```
        # Calculate and display dosage
        print("\n=== Treatment Recommendation ===")
        if age_months > 6:
            print(f"For patient >6 months old ({age_months/12:.1f} years):")
        else:
            print(f"For infant ≤6 months old ({age_months} months):")

        calculate_azithromycin(total_dose, total_day, times_per_day, double_first_day)

        # Print important notes
        print("\n=== Important Notes ===")
        print("- Take the medication at the same time each day")
        if double_first_day:
            print("- Double dose on first day only")
        print("- Complete full course even if symptoms improve")
        print("- Watch for gastrointestinal side effects")

        # Disclaimer
        print("\n=== DISCLAIMER ===")
        print("This tool is for educational purposes only.")
        print("It does not replace clinical judgment.")
        print("Always consult with a physician for medical decisions.")

    except ValueError:
        print("\nError: Please enter valid numeric values for weight and age.")

# Run the program
if __name__ == "__main__":
    pertussis_treatment()
```